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Global instability of the interaction between an oblique shock and a laminar boundary layer

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ABSTRACT

This study investigates the oblique shock-wave/boundary-layer interaction with a Mach number of 2.15 and a Reynolds number of 1×10^5. Both global stability analysis and direct numerical simulation are used to reveal the global instability characteristics and three-dimensional details of the incident shock flow. The results of global stability analysis indicate that stationary global instability occurs when the shock angle exceeds the critical angle 31.8°. At a shock angle equal to 33°, an additional unstable mode appears, which is oscillatory at large wavelength and gradually dominant when the wavelength decreases. As the wavelength is further reduced, the mode and its conjugation evolve into two stationary modes with different growth rates. A global instability criterion for incident shock flow is established based on the triple deck theory, which determines the instability only through free-stream conditions and shock angles. A direct numerical simulation is performed for the 32° shock angle case. It is found that secondary separation occurs during the nonlinear growth, which is absent in the two-dimensional base flow. Moreover, the separated flow undergoes a secondary perturbation growth, during which the dominant spanwise wavelength is doubled and the flow structures change significantly. The flow oscillates around a quasi-steady state in the end, indicating that a stationary unstable mode can develop unsteadiness without external disturbances.

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I. INTRODUCTION

The stability and transition of a laminar boundary layer have been under continuous investigation for over a century and have attracted increasing interest in recent decades due to their significant value in both fundamentals of flow and applications in engineering.[1] For instance, the stability and transition are related to the skin friction drag, aero-heating load, and the inlet flow quality of an aircraft. It is widely recognized that in a small disturbance environment, the paths of transition are typically divided into the following stages: (a) receptivity, (b) linear or eigenmode growth and transient growth, and (c) nonlinear growth and breakdown.[2] However, the physical mechanisms of transition processes have not been fully understood due to their complexity. Moreover, during high-speed flights, the shock-wave/boundary-layer interactions (SWBLIs)[3] are commonly present on engine inlets, control surfaces, transonic airfoils, etc.[4] This makes the transition mechanisms of high-speed flow more elusive, since SWBLIs significantly alter the characteristics of stability and dynamics of the flow, by introducing flow separation and wave system.

The canonical configurations of SWBLI consist of oblique shock wave incidence, compression ramp, double cone, double-fin, etc.[5] Taking the shock wave incidence case as an example, if the incident shock wave is strong enough, it can induce an adverse pressure gradient that is sufficient to cause boundary layer separation. The separation bubble has a significant impact on flow stability, as it can amplify upstream perturbations on the one hand, and has inherent instability on the other. In particular, when the intensity of the shock wave is just strong enough to cause incipient separation, the flow remains globally stable, exhibiting no inherent instability. At this point, separation bubbles can amplify upstream perturbations through convective mechanisms (e.g., Mack's first and second modes,[6] Görtler vortices,[7] and transient growth[8]) When the incident shock becomes stronger, the separation bubble develops three-dimensionality and the flow becomes globally unstable. Meanwhile, upstream perturbations are present in the actual flow. Hence, global instability and convective instability coexist in the flow, which makes the mechanism of transition more complex and requiring further investigation. Regardless of the perturbation growth pattern, the flow eventually undergoes nonlinear growth and transitions to turbulence typically in the form of the breakdown of streaks.

The instability caused by SWBLI not only induces boundary layer transition but also leads to low-frequency oscillation of the shock wave

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system[9] and "breathing" of the separation bubble, which may result in fatigue damage to the vehicle.[10,11] Therefore, extensive studies have been carried out on SWBLI, and various control methods have emerged in recent years.[12–14] Liepmann[15] and Ackeret *et al.*[16] were among the first to study the interaction between a shock wave and a boundary layer at transonic and low supersonic speeds experimentally, followed by the experimental observation of separation bubbles caused by the interaction with an oblique shock incident on the laminar boundary layer.[17,18] Thereby, the basic flow structure of SWBLI was clear and a large number of experiments have been conducted.[19–21] Meanwhile, the free interaction theory proposed by Chapman *et al.*[22] suggested that the initial separation process was independent of the downstream conditions, especially the shock wave intensity. Stewartson and Williams[23] established the triple deck theory based on the assumption of Lighthill[24] that the interaction region can be divided into an inviscid and irrotational outer layer, a rotational but inviscid middle layer, and a viscous bottom layer. Katzer[25] carried out two-dimensional (2D) numerical simulations for the interaction between an oblique shock and a laminar boundary layer with Mach numbers ranging from 1.4 to 3.4 and Reynolds numbers ranging from 1×10^5 to 6×10^6, confirming the precision of pressure scaling of triple deck theory.

Most of the research mentioned above has mainly focused on the measurement of steady aerodynamic properties. It was not until the late 1990s that the dynamic analysis of SWBLIs began to be systematically studied.[26] With the development of experimental measurement techniques and direct numerical simulation (DNS)/large eddy simulation (LES), a more comprehensive understanding of SWBLI mechanisms has been achieved. Ganapathisubramani *et al.*[27] investigated the effect of the upstream boundary layer on low-frequency instability in the compression corner (CC) interaction with a Mach number of 2 using high-speed particle image velocimetry. Sandham *et al.*[28] performed experiments and DNS on a transitional incident SWBLI with a Mach number of 6 and stated that transition develops from the Mack second mode instability superimposed on the downstream streaks. Giepman *et al.*[29] utilized high-resolution particle image velocimetry to investigate the three-dimensional (3D) instability and transition process of an oblique shock wave interaction. In a subsequent work,[30] a parametric study on this configuration was conducted, with a series of experiments for Mach numbers ranging from 1.6 to 2.3 and Reynolds numbers ranging from 1.4×10^6 to 3.5×10^6. They quantified the interaction strength using pressure rise and found that the boundary layer remains laminar for pressure rise ratios (p_2/p_1) less than 1.2. Fu *et al.*[31] employed DNS and wall-modelled LES to study laminar SWBLI of oblique shock wave incidence with a Mach number of 6. They discovered that the boundary layer can transition to turbulence without introducing any perturbation when the oblique shock is sufficiently strong. Xie *et al.*[32] investigated the characteristics of SWBLI in expansion–compression corner flow with a Mach number of 2.9 and various Reynolds numbers using DNS. In recent experimental studies, Chang *et al.*[33] observed the transition process of a hypersonic shock wave interacting with a flat plate flow by measuring the heat flux transfer. Butler and Laurence[34] investigated the convective instability (Mack's second mode) over a slender cone/flare with a Mach number of 6. Mahalingesh *et al.*[35] explored the effects of primary and secondary instabilities on the transition of SWBLI.

Various methods based on linear stability theory (LST) are also powerful tools for giving insight into flow stability. For instance, Balakumar *et al.*[36] performed LST and numerical simulations to investigate the second mode instability near the separation bubble for the experimental condition of Hyper-X (Mach number of 5.373, wedge angle of 5.5°). Niessen *et al.*[37] used LST to determine the contribution terms of the linear convective instability mechanisms in the laminar SWBLI. Yao *et al.*[38] investigated the dominant unstable modes of SWBLI at different Mach numbers using both LST and parabolized stability equations (PSE), and found that PSE provided better results. However, the assumptions of LST and PSE, such as the parallel and the slow-evolution, respectively, limit them to convective instability analysis. In 2007, Robinet[39] was the first to apply global stability analysis (GSA) to an oblique shock incidence (SI) interaction and found that the flow exhibited global instability when the shock angle exceeded a certain critical value, which was confirmed by DNS results. The GSA used by Robinet[39] is a two-dimensional global stability analysis (bi-global), which assumes the periodicity of perturbations in the spanwise direction, transforming the stability problem into a two-dimensional eigenvalue problem, as discussed in Sec. II B.

Since then, there has been an increasing amount of research conducted using GSA. Hildebrand *et al.*[40] obtained similar results to Robinet[39] at Mach 5.92, and employed adjoint global mode analysis to further analyze the instability mechanisms. Sidhart *et al.*[41] determined the critical deflection angle for global stability of a double wedge. In 2021, Hao *et al.*[42] analyzed the global stability of compression corner at Mach 7.7 with varying compression angles and wall temperatures, and established the global stability criterion for compression corners and double wedges according to the triple deck theory. They extended the criterion to axisymmetric double cones in 2022.[43] Additionally, Cao *et al.*[44] employed DNS to capture the transition process of the compression corner, and attributed the transition to the breakdown of streamwise counter-rotating vortices and streaks induced by the intrinsic instability of the flow through GSA. In 2023, Cao *et al.*[45] rounded the compression corner to controlled the separation, thus suppressing the intrinsic instability, and investigated the developments of globally stable and unstable flow under external disturbances by combining GSA, resolvant analysis, and DNS.

In the present study, we investigate the oblique SWBLI with a Mach number of 2.15 and a Reynolds number of 1×10^5, with a focus on the global instability characteristics. This study has three main objectives. The first is to determine the critical shock angle for global instability and the development of unstable modes at various shock angles by GSA. Although similar work has been done by Robinet,[39] it is expected that richer results can be obtained with the improvement of computational resources. The second is to establish a criterion to determine the global instability of the incident shock flow solely through free-stream conditions, following the approach taken for compression corner flows.[42] Finally, we attempt to answer the question of whether a stationary mode can develop unsteadiness, which is achieved by DNS.

The remainder of this paper is organized as follows: Sec. II introduces the numerical methods, including the flow simulation solver and GSA. Section III provides the geometric configuration, flow conditions, grid, and other computational details. The results and discussions are presented in Sec. IV. Finally, the main conclusions are summarized in Sec. V.

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II. NUMERICAL METHODS

A. Governing equations

The governing equations for numerical methods are the compressible Navier–Stokes equations, expressed in tensor form as

$$\frac{\partial \boldsymbol{U}}{\partial t}+\frac{\partial \boldsymbol{F}_j}{\partial x_j}=\frac{\partial \boldsymbol{F}_{v,j}}{\partial x_j}, \tag{1}$$

where $\boldsymbol{U}=(\rho,\rho u_i,\rho e)^{\mathrm{T}}$ is the vector of conservative variables, and

$$\boldsymbol{F}_j=\begin{pmatrix}\rho u_j\\ \rho u_i u_j+p\delta_{ij}\\ (\rho e+p)u_j\end{pmatrix},\quad \boldsymbol{F}_{v,j}=\begin{pmatrix}0\\ \tau_{ij}\\ \tau_{ij}u_i+q_j\end{pmatrix} \tag{2}$$

are the vectors of inviscid and viscous fluxes, respectively. Specifically, ρ is the density, u_i is the three components of velocity, p is the pressure, e is the total energy, δ_{ij} represents the Kronecker delta, τ_{ij} is the shear stress tensor expressed by Stokes' assumption,

$$\tau_{ij}=\mu\left(\frac{\partial u_j}{\partial x_i}+\frac{\partial u_i}{\partial x_j}\right)-\frac{2}{3}\mu\frac{\partial u_k}{\partial x_k}\delta_{ij}, \tag{3}$$

and q_j is the heat transfer given by Fourier's law as

$$q_j=-\kappa\frac{\partial T}{\partial x_j}. \tag{4}$$

To make Eq. (1) closed, the equation of state of a perfect gas is introduced,

$$p=\rho RT. \tag{5}$$

The specific heat ratio γ is 1.4, Prandtl number is 0.72, and the dynamic viscosity μ and thermal conductivity κ are calculated by Sutherland's equation.

The numerical simulations in this paper are performed using an in-house multi-block parallel finite-volume solver called PHAROS.[42,44,46,47] The inviscid fluxes are calculated by an approximate Riemann solver, Harten–Lax–van Leer–Contact scheme,[48,49] and the viscous fluxes are calculated by the second-order central difference. The implicit line relaxation method[50] is employed for the time iteration.

B. Global instability analysis

The linear stability theory decomposes $\boldsymbol{U}$ into the two-dimensional steady solution called the base flow $\boldsymbol{U}_{2\mathrm{D}}$ and small amplitude perturbation $\boldsymbol{U}'$,

$$\boldsymbol{U}(x,y,z,t)=\boldsymbol{U}_{2\mathrm{D}}(x,y,z,t)+\boldsymbol{U}'(x,y,z,t). \tag{6}$$

By substituting Eq. (6) into Eq. (1) and neglecting the high-order terms, the linearized Navier–Stokes equations (LNS) are obtained. If the perturbation is assumed to be periodic in the spanwise direction z, U' can be expressed in the modal form as

$$\boldsymbol{U}'(x,y,z,t)=\hat{\boldsymbol{U}}(x,y)\exp\left[\mathrm{i}\frac{2\pi}{\lambda}z-\mathrm{i}(\omega_r+\mathrm{i}\omega_i)t\right], \tag{7}$$

where λ is the spanwise wavelength, ω_r and ω_i are the angular frequency and the growth rate, respectively. Note that a positive growth rate implies globally unstable flows. By substituting Eq. (7) into the LNS, the instability problem is reduced to an eigenvalue problem,

$$A(\lambda)\hat{\boldsymbol{U}}=-\mathrm{i}(\omega_r+\mathrm{i}\omega_i)\hat{\boldsymbol{U}}, \tag{8}$$

where A denotes the global Jacobian matrix. Based on the modal assumption in Eq. (7), the global stability problem considered in this paper is a bi-global temporal stability problem of a spanwise periodic perturbation. The eigenvalues of A are computed by the implicit restart Arnoldi method implemented in ARPACK[51] for a given λ. The linear system equations are solved by the lower–upper decomposition implemented in Super-LU.[52]

III. COMPUTATIONAL DETAILS

A. Configuration and flow condition

The typical configuration of a shock wave incident on a flat plate is shown in Fig. 1. A weak shock wave is generated from the leading edge due to the viscous interaction. When the shock wave impinges on the laminar boundary layer of the flat plate with an angle θ, a strong adverse pressure gradient is formed and thus the laminar boundary layer is separated. The flow deflection caused by the separation bubble results in the formation of separation and reattachment shocks and a series of expansion waves.

The flow conditions are taken from the case studied by Degrez *et al.*[53] The free-stream conditions are as follows: $M_\infty=2.15$, $\rho_\infty=0.0246\,\mathrm{kg/m^3}$, $P_\infty=1082\,\mathrm{Pa}$, $T_\infty=153.33\,\mathrm{K}$, and $Re_\infty=1.25\times10^6\,\mathrm{m^{-1}}$. The inviscid shock incident position is fixed at 80 mm from the leading edge, resulting in the Reynolds number of $Re_{sh}=1\times10^5$ at this position. In viscous flow, the incident position is shifted upstream due to the thickening and separation of the boundary layer induced by the shock wave. According to Robinet's[39] stability study of this configuration, the shock angle θ in this study ranges from 31.5° to 33°.

B. Boundary condition and grid

The shock incidence is achieved by setting different inflow conditions in segments. Below the location where the shock wave enters the computational domain, the free-stream condition described in Sec. III A is applied, while above it, including the upper boundary, the flow parameters behind the shock obtained by the oblique shock relations are applied. The outflow condition at the right boundary is implemented by simple extrapolation. The wall is set to be no-slip and adiabatic with $T_w=273.6\,\mathrm{K}$. In the 3D DNS, the boundary condition in the spanwise direction is set as periodic. Three grids are generated for grid independent verification of 2D simulation and GSA (see more details in the Appendix). The computational grid of the base flow has

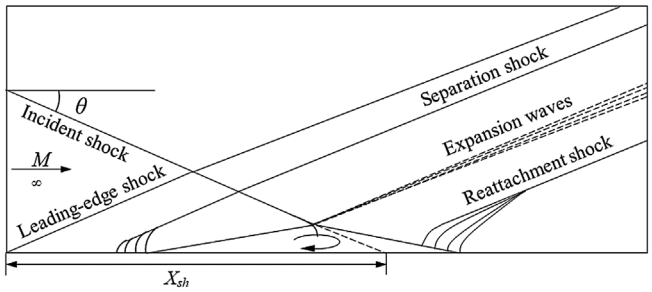


FIG. 1. Schematic of an oblique shock incident on a flat plate.

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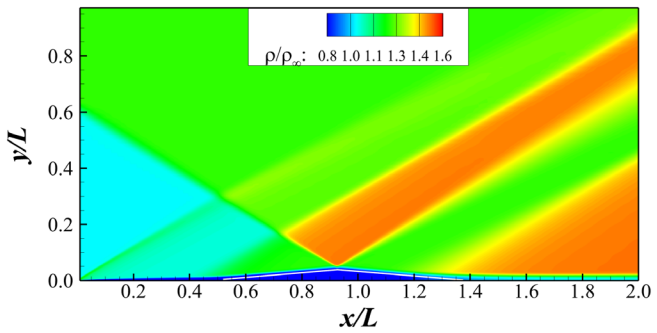


FIG. 2. Visualization of the flow structure at $\theta = 32^\circ$ (the white line represents the separation region).

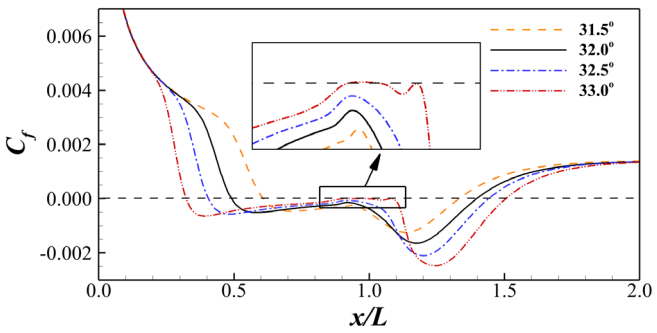


FIG. 3. Distributions of skin friction coefficient C_f at different shock angles.

600 × 300 cells in the streamwise and normal directions. The streamwise grid is appropriately refined near the separation region to capture the separation bubble accurately. The height of the first layer of the normal grid is 1×10^{-6} m, resulting in the grid Reynolds number on the order of 1. In addition, the normal grid is specially adjusted to ensure that shock impinges on the flat plate exactly at $x = 80$ mm for different shock angles.

IV. RESULTS

A. Base flow

The numerical simulations of 2D base flow have been conducted for shock angles of 31.5°, 32°, 32.5°, and 33°, respectively. The flow structure of the case with a 32° shock angle is visualized in Fig. 2, where the inviscid shock incident position $X_{sh} = 80$ mm is taken as the characteristic length L. The separation and reattachment points are located at $x/L = 0.5$ and $x/L = 1.4$, respectively. Within this region, there is a narrow separation bubble indicated by the white line.

The variations of skin friction coefficient C_f with x axis at different shock angles are shown in Fig. 3. The intensity of the shock wave increases with the shock angle, leading to an enlargement of the separation zone. A closer inspection of the magnified image shows that the skin friction coefficient distribution at $\theta = 33^\circ$ has crossed the line of $C_f = 0$, indicating that secondary separation occurs at $\theta = 33^\circ$. Figure 4 displays the density contours and the streamlines in separation region for different shock angles. The vortex core is consistently located at the rear of the separation bubble and gradually deforms with the shock angle, ultimately splitting into two parts at $\theta = 33^\circ$.

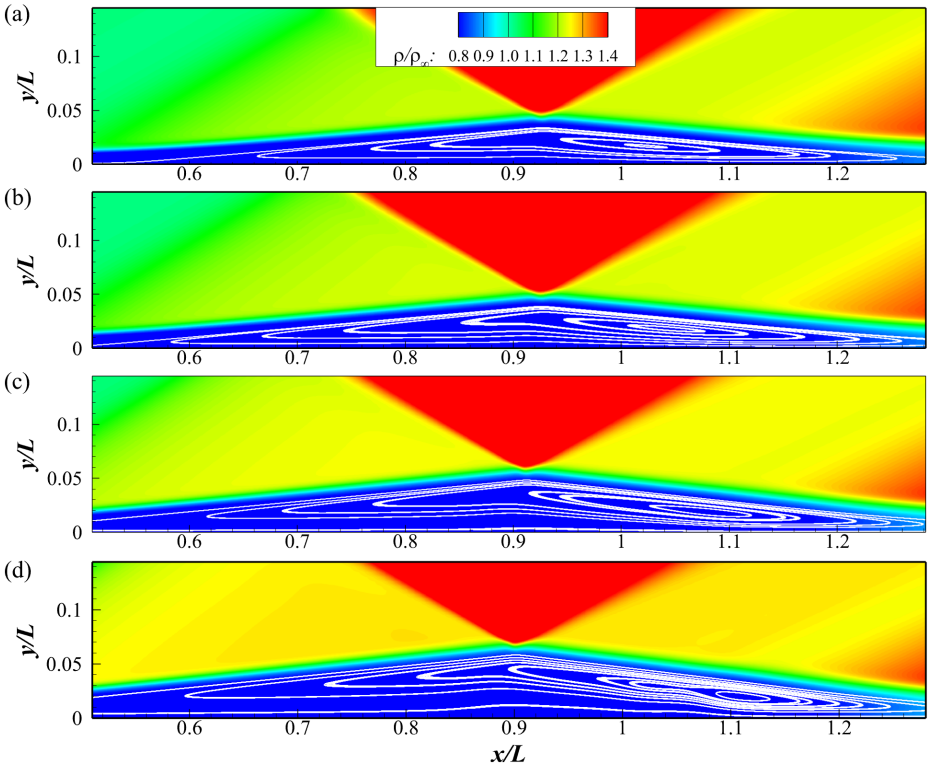


FIG. 4. Density contours and the streamlines in the separation region at (a) $\theta = 31.5^\circ$, (b) $\theta = 32^\circ$, (c) $\theta = 32.5^\circ$, and (d) $\theta = 33^\circ$.

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B. Global stability analysis

In order to obtain the global stability characteristics of the laminar boundary layer interacting with the oblique shock wave, GSAs are performed at various shock angles based on the base flows presented in Sec. IV A. The onset of global instability is first observed at a shock angle of 31.8°, consistent with the results of Robinet.[39] However, in the present study, the flow is found to be globally unstable at the dimensionless wavelength of $\lambda/L = 0.4$, but stable at $\lambda/L = 0.8$, in contrast to Robinet's results. As shown in Fig. 5, at $\lambda/L = 0.8$, all the growth rates ω_i are negative, indicating global stability of the flow; at $\lambda/L = 0.4$, one mode exhibits positive growth rate ω_i, signifying global instability of the flow.

The variations of the spanwise velocity growth rate ω_i and angular frequency ω_r with the wavelength λ are displayed in Fig. 6. For shock angles θ less than 31.8°, such as 31.5°, the boundary layer remains globally stable at any wavelength. When the angle exceeds the critical angle 31.8°, global instability occurs with a maximum growth rate around $\lambda/L = 0.33$. As the shock angle continues to increase, the maximum growth rate and the range of wavelength corresponding to the unstable state are larger. It is worth noting that when $\theta = 33°$, the boundary layer is highly unstable, such that in addition to the stationary mode (mode 1) observed in other shock angle cases, another unstable mode (mode 2) emerges, which is oscillatory at a large wavelength. When $\lambda/L = 0.27$, the growth rate of mode 2 exceeds that of mode 1, implying that the mode 2 becomes dominant. As the wavelength decreases, the frequency of mode 2 drops gradually and reaches zero at $\lambda/L = 0.18$ [see Fig. 6(b)]. It means that the mode 2 transforms into a stationary state, indicated as the two branches evolving from mode 2. The evolution of mode 1 and mode 2 with wavelengths at $\theta = 33°$ is shown in Fig. 7, where the direction of the arrow represents a decrease in the dimensionless wavelength λ/L from 0.28 to 0.18. With the decrease in wavelength, mode 2 and its conjugate gradually approach, i.e., the angular frequency decreases. As the angular frequency decreases to zero, the two conjugate solutions collide and then transform into two stationary modes with different growth rates (represented as solid green symbols). It is consistent with the two branches in the growth rate of mode 2 when $\lambda/L < 0.18$.

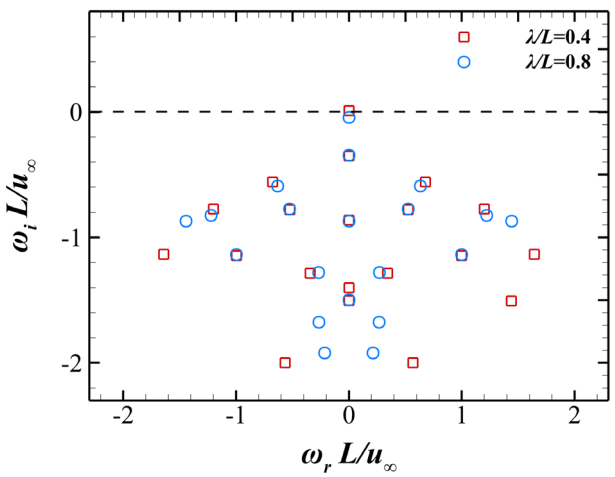


FIG. 5. Eigenvalue spectra at $\theta = 31.8°$ with $\lambda/L = 0.4$ and $\lambda/L = 0.8$.

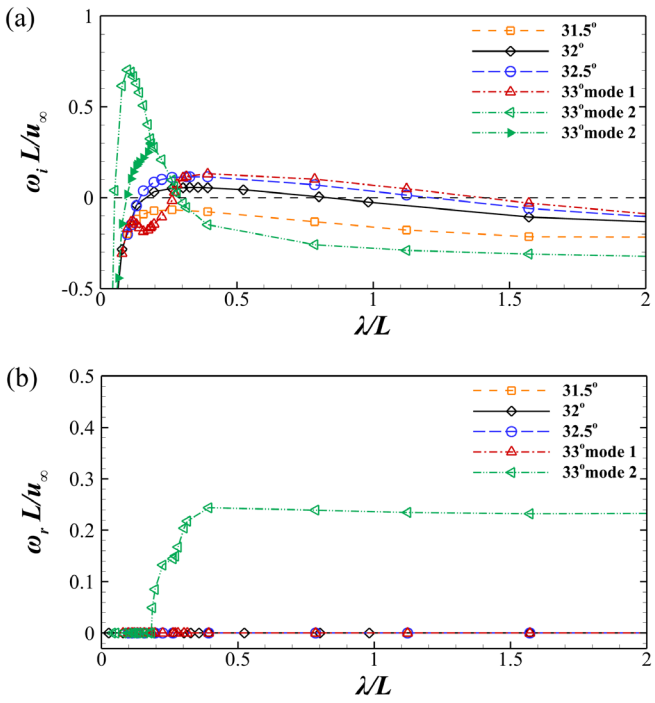


FIG. 6. Variations of dimensionless (a) growth rates and (b) angular frequencies of the most unstable modes with wavelengths at various shock angles.

Figure 8 displays the contours of the real part of spanwise velocity for the most unstable mode at various shock angles. Note that Figs. 8(c) and 8(d) are both for $\theta = 33°$, but represent mode 1 and mode 2, respectively. The spanwise velocity perturbations are all concentrated in the separation zone, with the maximum perturbation located at the rear of the separation zone. When $\theta = 33°$, the spanwise velocity of mode 1 appears flocculent diffusion [Fig. 8(c)], which is speculated to be related to the splitting of the separation bubble in Fig. 4(d). For

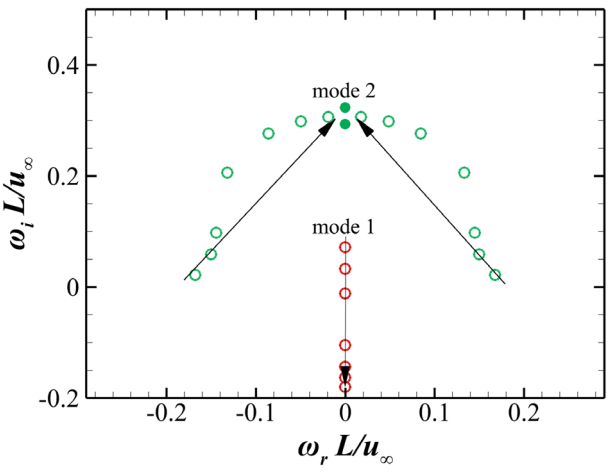


FIG. 7. The evolution of the mode 1 (red symbols) and mode 2 (green symbols) with wavelengths at $\theta = 33°$.

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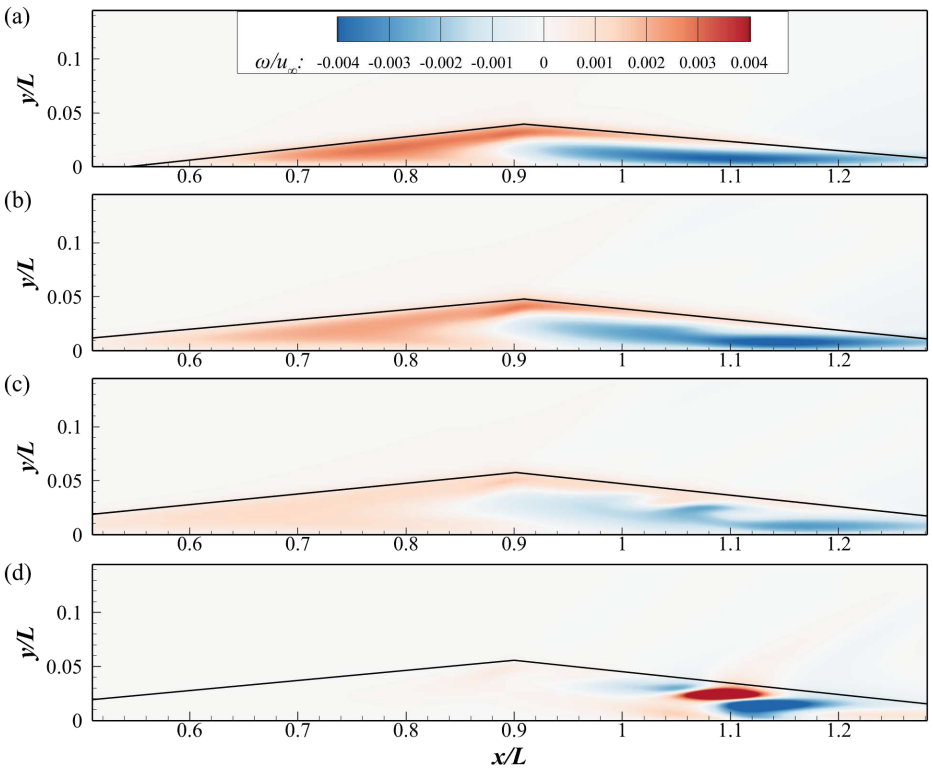


FIG. 8. Real part contours of spanwise velocity of the most unstable modes at different shock angles (black line represents separation region): (a) $\theta = 32^\circ$, $\lambda/L = 0.3$; (b) $\theta = 32.5^\circ$, $\lambda/L = 0.3$; (c) $\theta = 33^\circ$, $\lambda/L = 0.4$ (mode 1); and (d) $\theta = 33^\circ$, $\lambda/L = 0.1$ (mode 2).

mode 2 [Fig. 8(d)], the wavelength is smaller and the spanwise velocity perturbation is larger, which is consistent with the higher growth rate of mode 2 in Fig. 6.

C. Global instability criterion using triple deck theory

Triple deck theory[23] is an asymptotic model to describe boundary layer separation based on Lighthill's[24] assumption that the boundary layer can be divided into three parts, or decks, with three distinct characteristics. The upper deck is inviscid and irrotational; the middle deck is inviscid but rotational; the lower deck is viscous and can be solved by an incompressible boundary layer equation. The most important contribution of triple deck theory for this paper is that it provides a scaled ramp angle α^* to quantify the interaction intensity in SWBLI. Assuming a compression corner flow with an actual ramp angle of α, a Reynolds number at the corner of Re_L and the free-stream Mach number of M_∞, the scaled ramp angle α^* is

$$\alpha^* = \frac{\alpha Re_L^{1/4}}{C^{1/4} 0.332^{1/2} \left(M_\infty^2 - 1\right)^{1/4}}, \quad C = \frac{\mu_w T_\infty}{\mu_\infty T_w}, \tag{9}$$

where C is the Chapman–Rubesin parameter.

Hao *et al.*[42] established a global stability criterion for compression corner boundary layer based on the scaled ramp angle α^*. The critical range of the scaled ramp angle for the global instability of compression corner flows is $3.44 < \alpha^* < 4.59$, corresponding to the wall temperature from low to high. This paper attempts to extend the global instability criterion to incident shock flows in terms of α^*. First, the shock angle θ is converted to a ramp angle α of compression corner with the same pressure rise using the oblique shock relations. Then, GSAs are performed for different flow conditions to find the global instability boundary. The flow conditions are selected from recent numerical and experimental studies, as listed in Table I.

The critical scaled ramp angles are shown in Fig. 9, which also includes the criterion for compression corner flows[42] as a comparison. It is worth noting that the variation of the critical angles is governed by the wall-to-free-stream temperature ratio, rather than the wall-to-total temperature ratio used by Hao *et al.*,[42] where the cases share the same Mach number. In this paper, since the cases have different Mach numbers, the wall-to-free-stream temperature ratio is more appropriate to characterize the temperature effect. The instability boundary for shock incidence is $3.5 < \alpha^* < 5.5$. It has been observed that both the incident shock and compression corner flows exhibit a similar trend in global instability concerning wall temperature. Specifically, The higher the wall temperature, the larger the critical ramp angle α^* is, indicating that the boundary is more stable with a higher wall temperature.

TABLE I. The flow conditions for the establish of global instability criterion of incident shock flow.

	M_∞	Re_L	T_w/T_∞	$\theta(^\circ)$	$\alpha(^\circ)$	α^*
Chang *et al.*[33]	7	1.28×10^6	3.2	11.7	10.08	4.26
Fu *et al.*[31]	6	2.7×10^6	4.5	12.6	8.72	4.47
Hildebrand *et al.*[40]	5.92	1.15×10^5	6.95	13.7	11.29	4.76
Present	2.15	1×10^5	1.93	31.8	9.93	3.97

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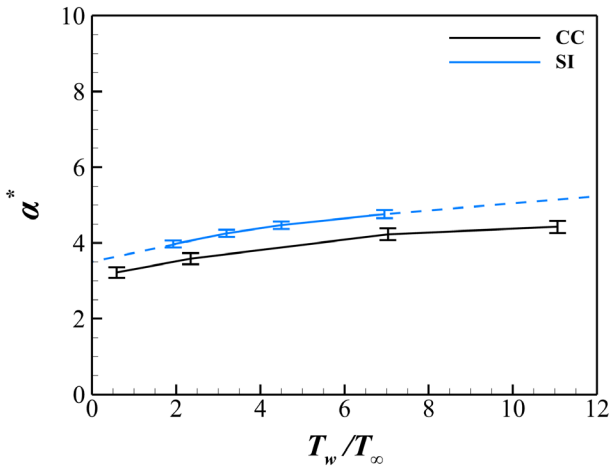


FIG. 9. The critical scaled angles α^* of global instability for shock incidence (SI) and compression corner (CC)[42] problems as a function of wall temperature.

Moreover, under the same flow conditions, the critical scaled ramp angles of incident shock flow are larger than those of the compression corner, which demonstrates that the incident shock flow is more stable.

D. Direct numerical simulation

The base flow characteristics and global stability of the incident shock flow have been revealed in Secs. IV A and IV B. In order to verify the results and to further obtain the three-dimensional features, a DNS for a shock angle of 32° is performed in this section. According to the results of GSA, only the stationary unstable mode is available at this condition and the growth rate is relatively small (see Fig. 6). To investigate whether a stationary unstable mode can develop an oscillatory unstable mode in the absence of external disturbances, it is important to examine the post-perturbation saturation phase. However, the linear growth phase is expected to be prolonged due to the low growth rate. To save computational resources, the most unstable mode obtained by GSA ($\lambda/L = 0.33$) is superimposed on the 2D base flow as the initial flow field of DNS, expressed in the following equation as

$$\boldsymbol{U}_{\text{initial,3D}} = \boldsymbol{U}_{\text{2D}} + A\boldsymbol{U}_{\text{GSA}}, \tag{10}$$

where A is the amplification factor of the GSA unstable mode, determined by setting the maximum spanwise velocity in the flow field equal to one-thousandth of the free-stream velocity,

$$Aw_{\max} = 1‰\, u_\infty. \tag{11}$$

In the present case, $A = 0.28$. The spanwise length of computational domain is twice the wavelength of the most unstable mode in GSA. The grid is created by replicating the 2D grid with $600 \times 300 \times 80$ cells in streamwise, normal, and spanwise directions, respectively. The DNS is also performed using PHAROS with the same numerical schemes described in Sec. II A, except that the time scheme is changed to a second-order implicit scheme.[54] The time step is set to 40 ns and the resolution has been verified with a smaller time step of 20 ns. The physical computational time is 81 ms, corresponding to a non-dimensional time of $tu_\infty/L = 540$. To facilitate a comparison with the GSA results, the nominal spanwise velocity at a specific streamwise location δ_w is defined as follows:

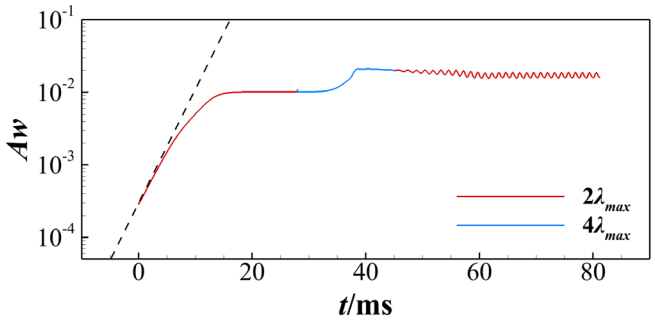


FIG. 10. The temporal history of σ_w at the $x/L = 1.04$ slice (the slope of the dashed line is the growth rate of GSA).

$$\delta_w = \sqrt{\frac{1}{N_y N_z}\sum_{j=1}^{N_y}\sum_{k=1}^{N_z}\left(\frac{w}{u_\infty}\right)^2}, \tag{12}$$

where N_y and N_z are the numbers of grid cells in the normal and spanwise directions, respectively. The temporal history of δ_w at the $x/L = 1.04$ slice is depicted as the red line in Fig. 10, where the dashed line represents the growth rate predicted by GSA and the blue line will be explained later. Because the initial condition is constructed using the most unstable mode, the flow develops directly from the late stage of linear growth. Although the linear growth stage is short, it can still be observed that the linear growth rates of DNS and GSA are very close. In fact, the growth rate of DNS is 6.3×10^{-4}, while that of GSA is 6.7×10^{-4}. At $t = 17$ ms, the spanwise velocity reaches saturation, but interestingly, it undergoes a secondary growth at $t = 32$ ms. After the secondary growth, the flow continues to develop and starts to oscillate at $t = 47$ ms. This answers the previously raised question that a stationary mode can indeed develop an oscillatory flow. The following discussion will delve into the details of the perturbation growth process and flow structures.

Figure 11 displays the spanwise velocity distribution of the xy and yz planes during the linear growth phase, specifically of the $z/L = 0.4$ and $x/L = 1.04$ slices at $t = 5$ ms. In the xy plane, the spanwise velocity distribution is identical to the eigenmode obtained from the GSA results presented in Fig. 8(a). In the yz plane, it can be observed that two pairs of perturbation waves are distributed in the spanwise direction, verifying the validity of the perturbation superposition in Eq. (10). To further analyze the flow, the xy plane at $y/L = 0.01$ is intercepted, indicated by the dashed line in Fig. 11. The spanwise velocity contours at different moments in this plane are displayed in Fig. 12, with black lines representing the separation and reattachment lines. It should be noted that the time intervals in Fig. 12 are not uniform, and shorter intervals are used for the nonlinear and oscillatory phases to provide a detailed flow field evolution. The entire process from 0 to 81 ms is shown as a movie in the multimedia file available online.

By comparing Figs. 10 and 12, the evolution of the flow field can be analyzed. In the linear growth phase (0–5 ms), the spanwise velocity

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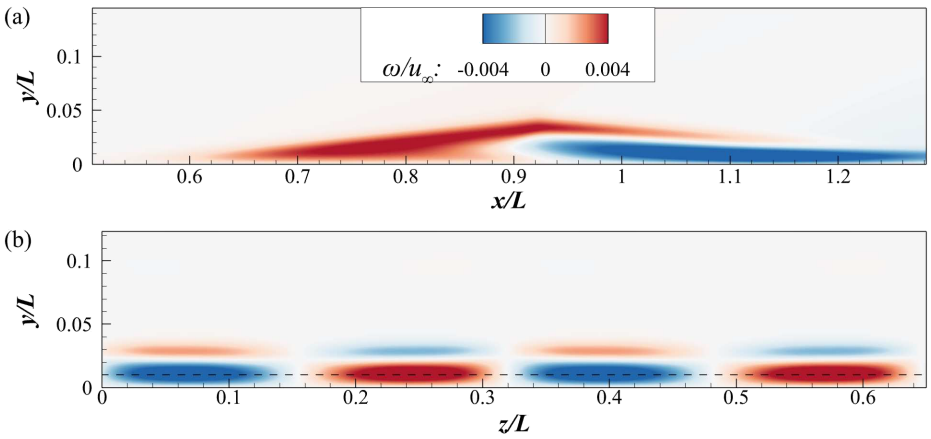


FIG. 11. The spanwise velocity contours of the $z/L = 0.4$ and $x/L = 1.04$ slices at $t = 5$ ms.

gradually grows. At the end of the linear growth [see Fig. 12(a)], the separation line is still straight but the reattachment line starts to twist slightly. In the nonlinear phase [5–17 ms, see Figs. 12(b) and 12(c)], the separation line appears to fluctuate and moves upstream slightly, while the reattachment line distorts into a wave-like shape; two new elliptical secondary separation bubbles appear, located in the middle of each pair of perturbation waves. After the nonlinear phase (17–32 ms), the first perturbation growth ends and the spanwise velocity saturates; during this phase [see Fig. 12(d)], the flow field structure exhibits minimal variation. Surprisingly, a secondary growth occurs (32–39 ms), characterized by a short growth and rapid saturation. Of particular interest is the accompanying changes in the flow field structure [see Figs. 12(e)–12(i)], where the perturbation waves located on the sides squeeze toward the middle, leading to the gradual merging of the two pairs into a single pair of waves; at the same time, the two secondary separation bubbles tend to tilt and migrate toward the middle, eventually coalescing into a large separation bubble. Consequently, significant transformations occur in the separation and reattachment lines. Specifically, the middle of the separation line shifts downstream, accompanied by the shrinking of the perturbation waves in the central region, while the two sides of the separation line move upstream along with the growth of perturbation wave, ultimately showing a bow shape configuration. The reattachment line transforms from a uniformly wavy shape into a mountain peak with two close peaks. After the secondary growth phase (39–60 ms), the perturbation becomes unsteady and ultimately oscillates around a reference value, achieving a quasi-steady state (60–81 ms). This is reflected in the flow field structure [see Figs. 12(j)–12(n)], where the spanwise velocity shows periodic oscillations mainly in the spanwise direction, accompanied by fluctuations in both separation and reattachment lines.

The results above illustrate that the sub-harmonic instability dominates the flow field through the secondary growth and eventually evolves into an oscillatory state. To confirm that there is no greater instability with longer wavelength, the spanwise length is extended to four times the wavelength of the most unstable mode in GSA results. To save computational resources, the flow field after the saturation of the primary perturbation growth ($t = 28$ ms) in the twice wavelength domain is replicated as the initial flow field of the new computational domain. The temporal history of the spanwise velocity is shown as the blue line in Fig. 10. The secondary growth progress of the computational domain with four times the wavelength is found to be identical to that with twice the wavelength. Figure 13 gives the distribution of spanwise velocity at the start and the end of the secondary growth for the new computational domain. It can be seen that the wavelength of the perturbation wave is doubled, and the shape of the separation and reattachment lines is similar to the result of the twice-wavelength computational domain. This observation confirms that the sub-harmonic wave with twice the wavelength of the most unstable mode in GSA is the dominant instability in this flow.

Figure 14 depicts the comprehensive flow structures in the steady or quasi-steady state after the primary and secondary perturbation growths. After the primary growth [Fig. 14(a)], secondary separation bubbles arise in the middle of each pair of perturbation waves. Due to the interplay between the secondary separation bubbles and the perturbation waves, the reattachment lines are distorted into a wave shape, and the streamlines at the crests and troughs are manifested as node (N) and saddle (Sa) points. Some streamlines that originate from the nodes on the reattachment line terminate at spiral (Sp) points that appear in pairs near the secondary separation line, while others terminate at the nodes on the separation line. Furthermore, the number of node and saddle points on the reattachment line is twice that on the separation line. This corresponds to the phenomenon that the doubling of streaks after reattachment in a fully developed perturbation flow,[55] and even higher harmonics have been reported in previous studies. Downstream of the flow, the spanwise velocity of the perturbation leads to the formation of two pairs of counter-rotating vortices, as shown in the streamlines of the x-slice. The streamwise vorticity is mainly concentrated in the boundary layer. After the secondary growth [Fig. 14(b)], the fusion of the two secondary separation bubbles results in the further alteration of singularity points in the flow field. The saddle points on the reattachment line vanish, replaced by a node and a spiral point. Similar to the results after the primary growth, some streamlines originating from the reattachment line partially terminate at the node on the separation line and partially terminate at the spiral point near the secondary separation line. The downstream counter-rotating vortices are deformed and halved in number, and streamwise vorticity remains confined inside the boundary layer.

The skin friction distributions of 2D and 3D results are compared in Fig. 15, where the 3D results are obtained by performing time- and spanwise-averaging over the time interval from 60 to 81 ms during the

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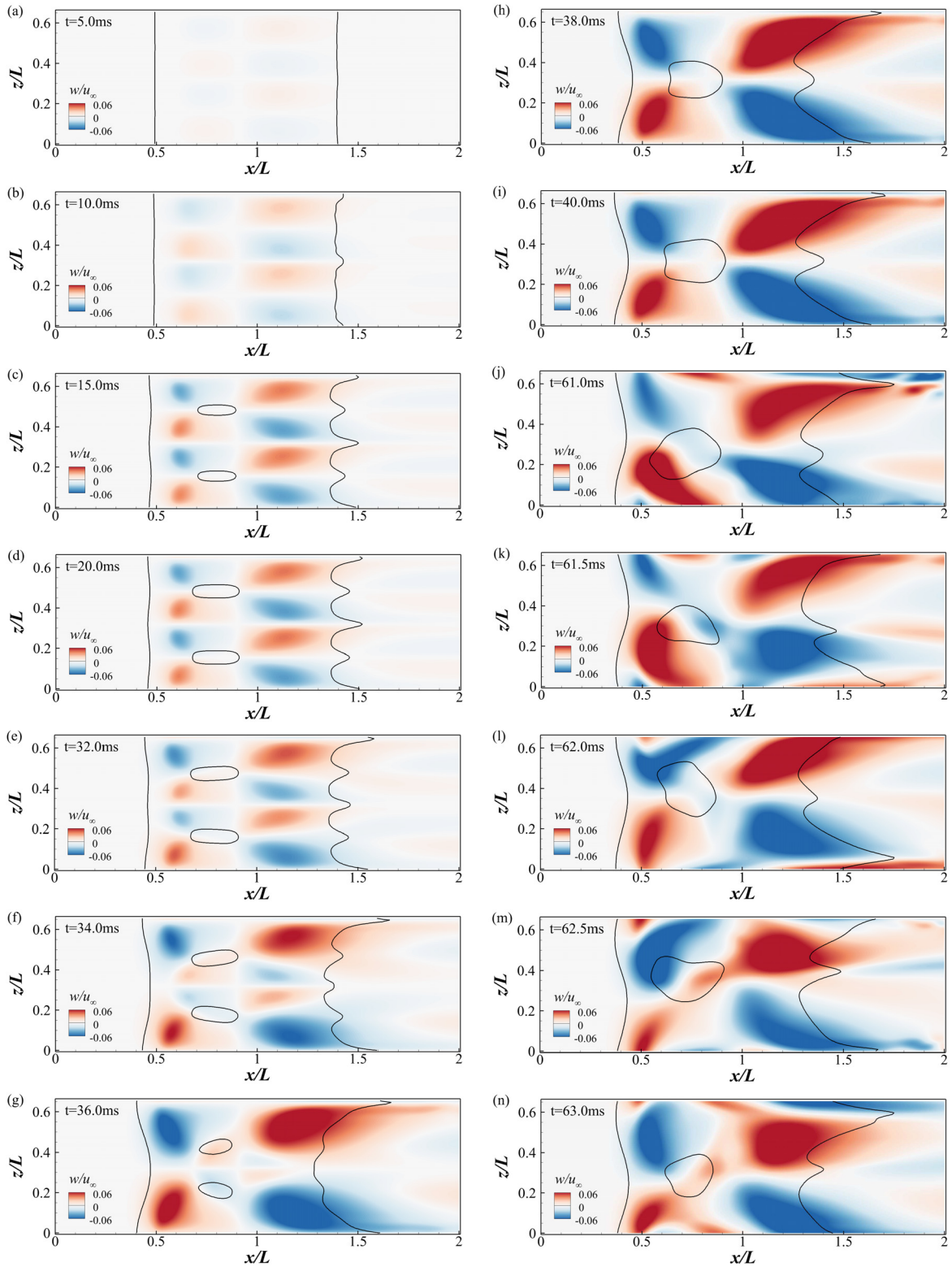
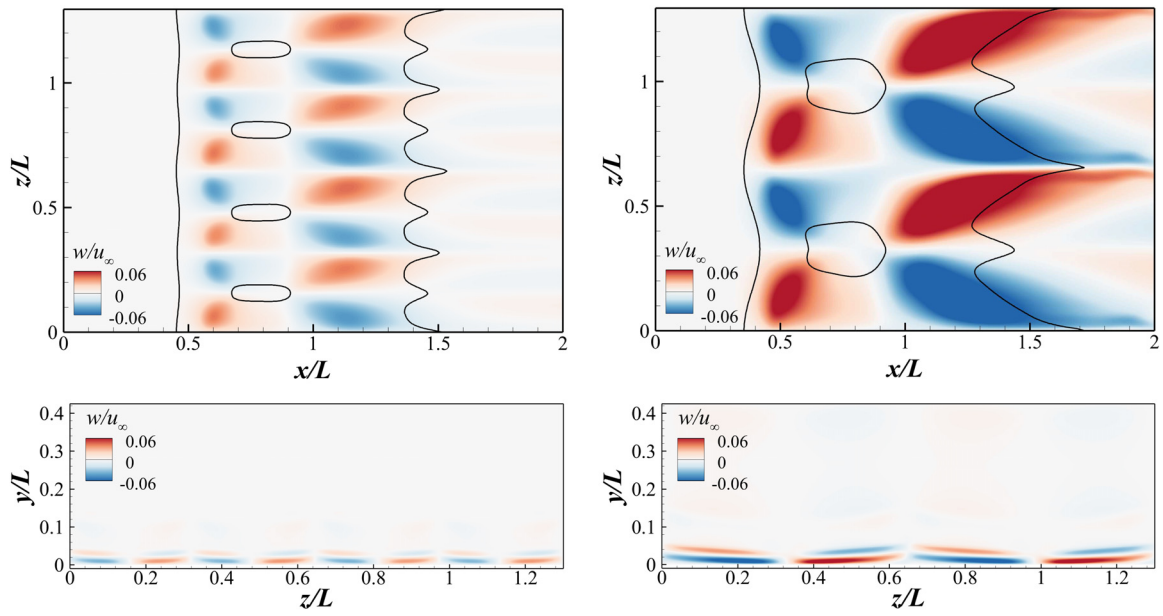


FIG. 12. The spanwise velocity contours at different moments at the $y/L = 0.01$ slice with the separation and reattachment lines superimposed. Multimedia available online.

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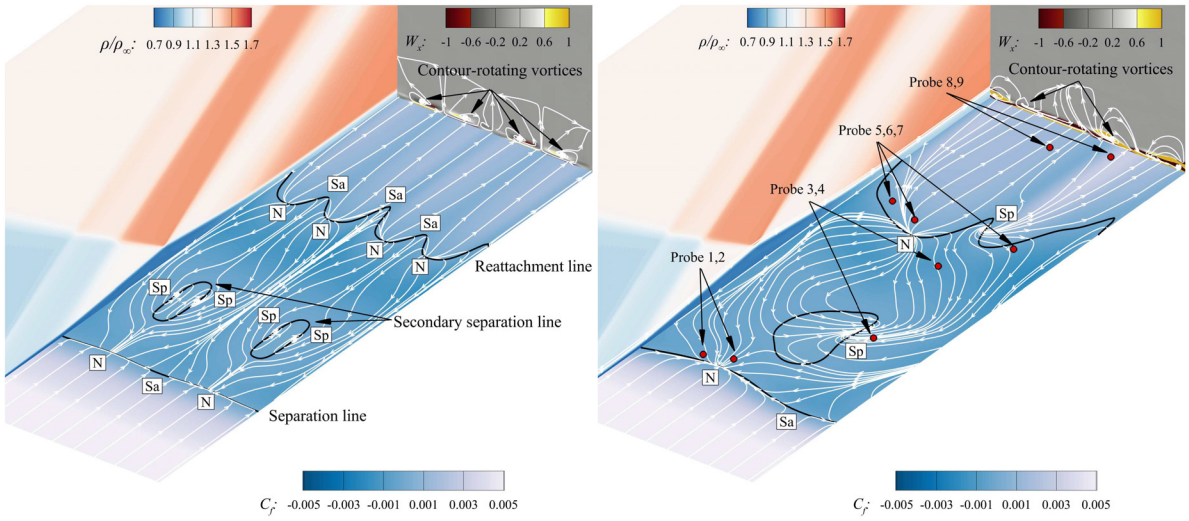


(a) $t = 29.5$ ms (b) $t = 45.5$ ms

FIG. 13. The spanwise velocity distribution at the $y/L = 0.01$ and $x/L = 1.04$ slices for spanwise length of four times the most unstable wavelength: (a) $t = 29.5$ ms and (b) $t = 45.5$ ms.

quasi-steady state. The three-dimensionality of the flow extends the separation zone. Specifically, the 3D averaged separation point significantly shifts upstream compared to the base flow, which is consistent with the variation in separation line observed in Fig. 12. However, the average reattachment point remains almost unchanged. The secondary separation bubble that appears in the 3D flow are not reflected in the skin friction distribution due to the spanwise-averaged.

Figure 16 presents the streamwise vorticity distribution for various x-slices at different moments, superimposed with in-plane streamlines. It can be seen from the streamlines that the counter-rotating vortices appear after reattachment, which is also observed by Cao *et al.*[44] In addition, the value of the streamwise vorticity starts to decrease near the reattachment region. By comparing Figs. 16(a) and 16(b), it is found that after the secondary growth, the streamwise



(a) $t = 25$ ms (after primary growth). (b) $t = 70$ ms (after secondary growth).

FIG. 14. Visualization of flow structure at different moments: (a) $t = 25$ ms (after primary growth) and (b) $t = 70$ ms (after secondary growth), composed of density contour, skin friction coefficient and downstream streamwise vorticity contours, superimposed on the streamlines of the wall and downstream.

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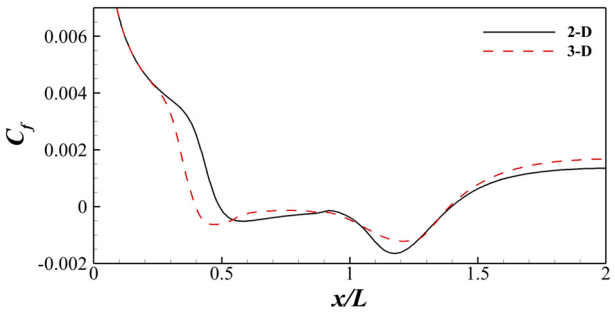


FIG. 15. The skin friction distributions of 2D flow and time- and spanwise-averaged 3D flow at $\theta = 32^\circ$ (averaged over the time interval from 60 to 81 ms).

vorticity is larger, and the counter-rotating vortices change from two pairs to one pair, consistent with the doubling of the wavelength that has been repeatedly mentioned before.

To record the fluctuations in the oscillation phase, nine probes are strategically placed in the flow field, as shown in Fig. 14(b). Probes 1 and 2 are located near the separation line; probes 3 and 4 are situated within the separation zone; probes 5, 6, and 7 are located near the reattachment line; and probes 8 and 9 are positioned downstream of the flow field after reattachment. The time interval for sampling is equivalent to the time step ($\Delta t = 40$ ns), with recording starting at $t = 60$ ms when the flow has reached a quasi-steady state. The recording lasts for more than 20 ms in total, and 530 000 sets of data are recorded. The Strouhal number (based on characteristic length L and free-stream velocity u_∞) of sampling frequency is $St = 3750$, which is sufficient to capture very high-frequency structures. Additionally, the recording duration is long enough to capture the low-frequency structures. The power spectral density (PSD) spectra are obtained through fast Fourier transform (FFT) and then are made dimensionless by the free-stream pressure and characteristic time as follows:

$$\mathrm{PSD} = \mathrm{FFT}[p(t)]L/u_\infty/{p_\infty}^2, \tag{13}$$

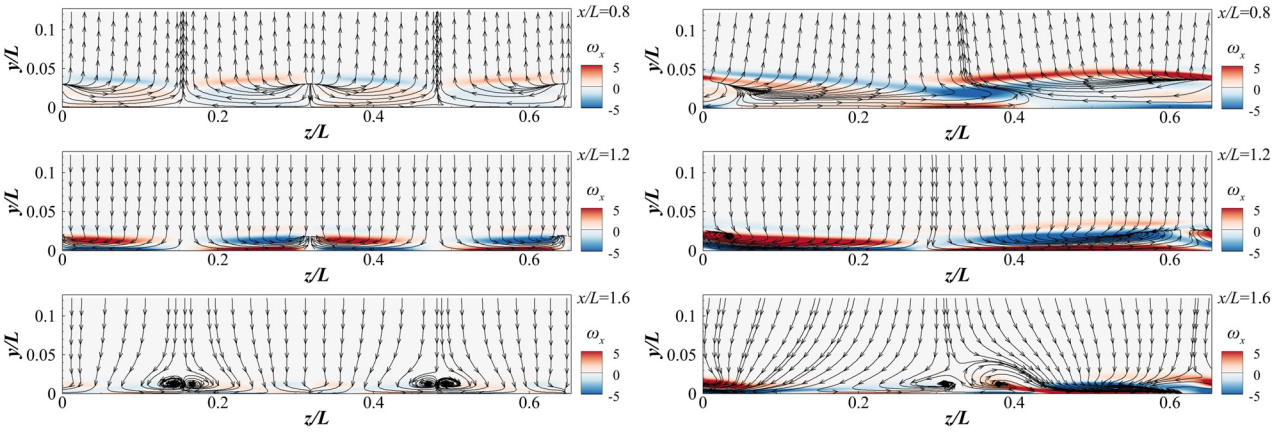
where $p(t)$ is the pressure recorded by the probes as a function of time. The PSD results are plotted in Fig. 17. It can be observed that at different positions, the pressure fluctuation exhibits an identical fundamental frequency of $St = 0.06$ approximately, which corresponds to $f = 420$ Hz. In addition, harmonics of this fundamental frequency are also present, including 800 Hz, 1200 Hz, and so on. However, high-frequency harmonics are no longer evident downstream and are replaced by broad-frequency fluctuation. The high-frequency fluctuation may interact with streaks, leading to the breakdown of streaks and triggering the onset of turbulence. Further investigation is needed to better understand this phenomenon.

V. CONCLUSION

This work presents a numerical investigation of the global instability of the oblique SWBLI with a Mach number of 2.15 and a Reynolds number of 1×10^5 for various shock angles.

The 2D base flow results indicate that the separation length increases with the shock angle, and the secondary separation occurs at $\theta = 33^\circ$, when the main separation bubble splits into two bubbles. The GSA results reveal that the global instability of the boundary layer first appears at $\theta = 31.8^\circ$ with the dimensionless wavelength $\lambda/L = 0.4$. As the shock angle increases, the boundary layer flow becomes more and more unstable, and the perturbation growth rate reaches a maximum at $\lambda/L = 0.33$. At a shock angle of $\theta = 33^\circ$, in addition to the stationary mode observed at smaller shock angles, another unstable mode oscillating at large wavelengths emerges and gradually dominates as the wavelength decreases. When the wavelength decreases to $\lambda/L = 0.18$, the oscillatory mode collides with its conjugate mode and bifurcates into two stationary modes with different growth rates.

Furthermore, a global instability criterion for incident shock flow is established based on the triple deck theory. The scaled ramp angle α^* is utilized to quantify the intensity of the interaction between shock wave and boundary layer, which can be obtained solely through free-stream conditions. The global instability boundary is determined to be $3.5 < \alpha^* < 5.5$, and α^* increases with wall temperature. This trend of



(a) $t = 25$ ms (after primary growth). (b) $t = 70$ ms (after secondary growth).

FIG. 16. Streamwise vorticity distributions for various x-slices at different moments superimposed with in-plane streamlines: (a) $t = 25$ ms (after primary growth); and (b) $t = 70$ ms (after secondary growth).

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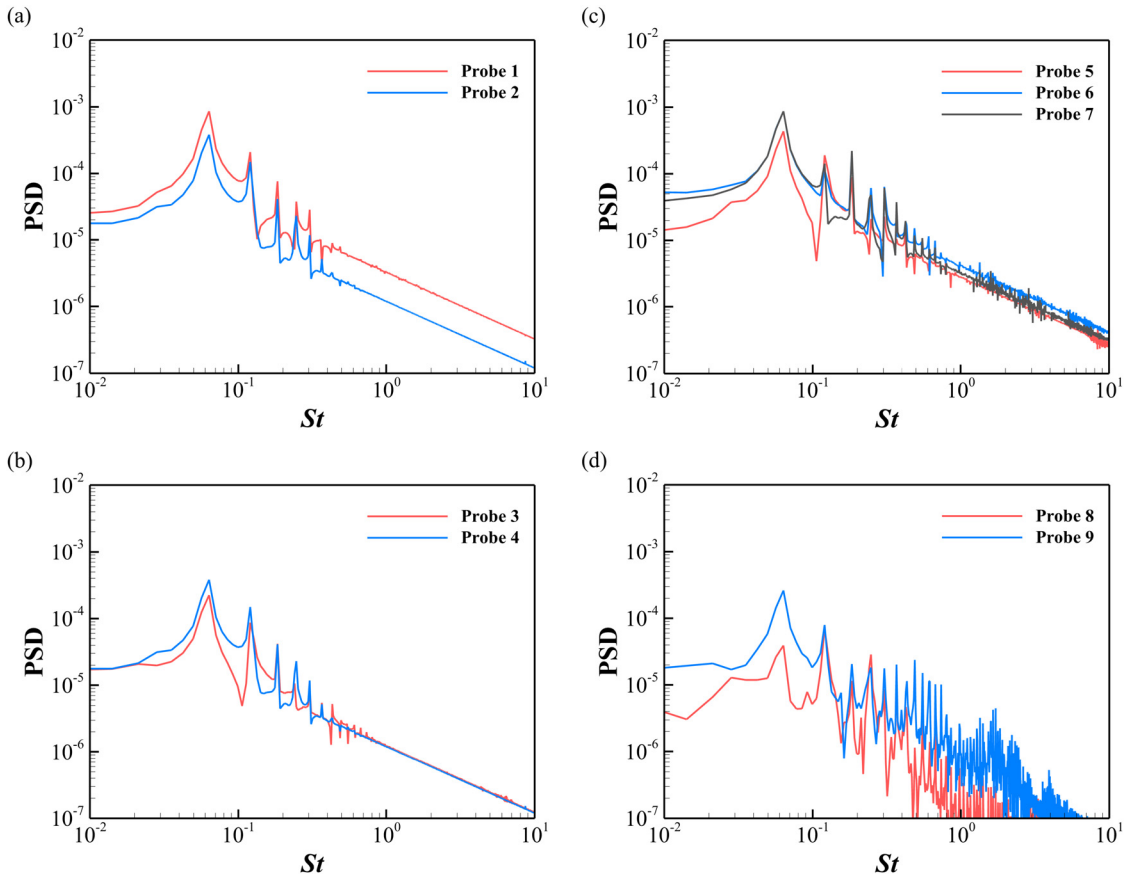


FIG. 17. The PSD results of pressure fluctuations at (a) probes 1–2 (separation line); (b) probes 3–4 (separation bubble); (c) probes 5–7 (reattachment line); and (d) probes 8–9 (downstream).

increasing stability with higher wall temperature is similar to that observed in compression corner flow.[42] However, for incident shock flow, the critical α^* is found to be larger than that of compression corner flow at the same flow conditions, indicating that the incident shock flow is more stable.

The DNS results not only verify the GSA predictions, but also reveal the flow evolution after the saturation of perturbation. A DNS is performed for a shock angle of 32°, which contains only one stationary unstable mode according to the GSA results. It is found that in the nonlinear growth phase, secondary separation appears, which is not present in the 2D base flow. After the primary perturbation growth which is consistent with the stationary mode found by GSA, the separation flow undergoes a secondary perturbation growth. During this period, the instability wavelength is doubled, and two secondary separation bubbles are merged into one. After the saturation of the secondary growth, the flow oscillates around a quasi-steady state. This indicates that a globally stationary unstable mode can develop unsteadiness in the absence of external disturbances.

The mechanism behind the secondary growth will be investigated in future work, and additional data will be collected to further validate the proposed global instability criterion.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Ziming Song: Conceptualization (equal); Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (equal); Writing – original draft (lead); Writing – review & editing (equal). **Jiaao Hao:** Conceptualization (equal); Methodology (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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APPENDIX: GRID INDEPENDENT VERIFICATION

The detailed parameters of coarse, medium, and fine girds are listed in Table II. The 2D skin friction coefficient distributions with $\theta = 32^\circ$ and the eigenvalue spectra of GSA with $\theta = 32^\circ$ and $\lambda/L = 0.33$ for different grids are compared in Figs. 18 and 19, respectively. In both 2D and GSA simulations, the results of medium and fine grids are very close, indicating that the medium grid has achieved grid-independence. Therefore, the medium grid is used for calculation in this paper.

TABLE II. The parameters of coarse, medium, and fine girds for grid independence verification.

Grid	Coarse	Medium	Fine
Streamwise cells	400	600	800
Normal cells	200	300	400
Normal cells in boundary layer	150	200	250
Grid Reynolds number	1.25	1.25	0.6

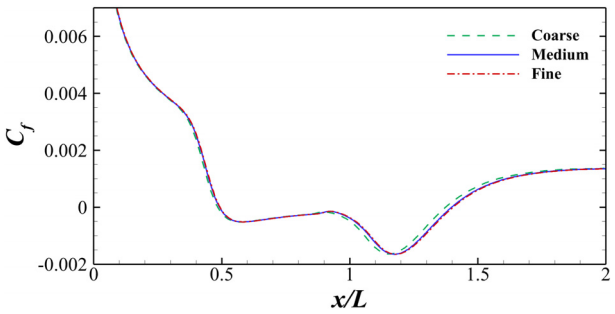


FIG. 18. The skin friction distributions at $\theta = 32^\circ$ obtained on different grids.

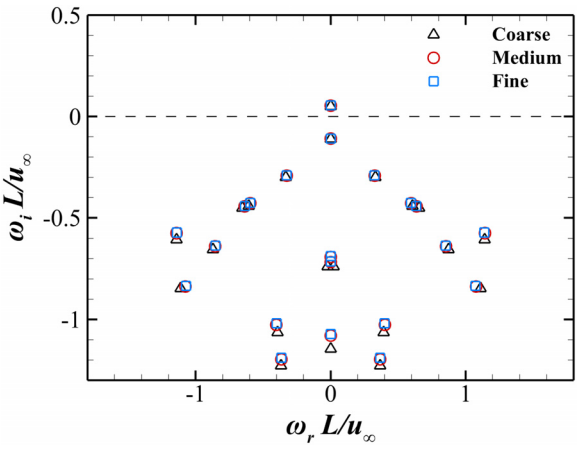


FIG. 19. Eigenvalue spectra at $\theta = 32^\circ$ and $\lambda/L = 0.33$ obtained on different grids.

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